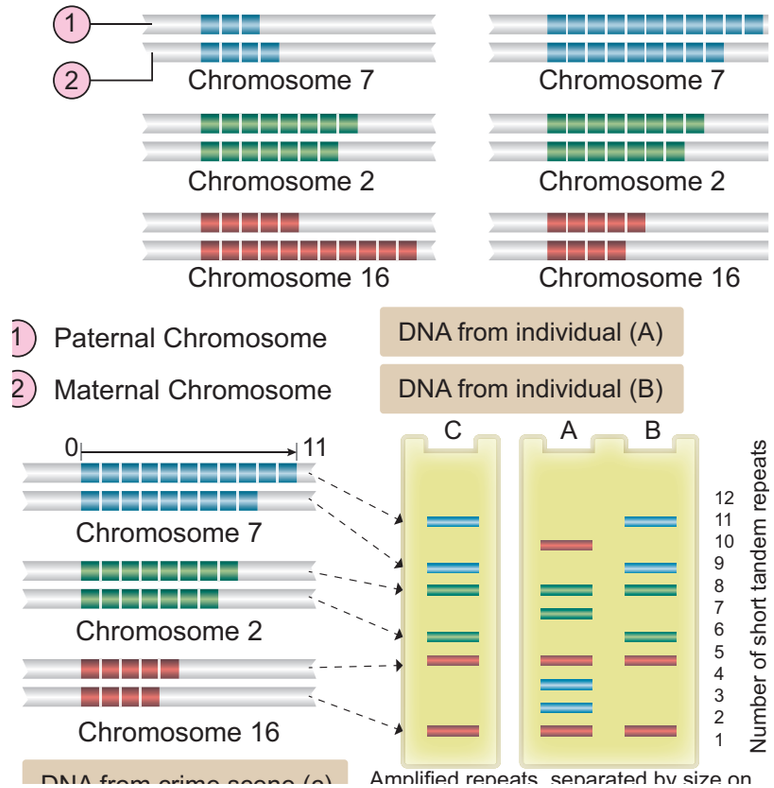


Fig. 5.15 Schematic representation of DNA fingerprinting : Few representative chromosomes have been shown to contain different copy number of VNTR

from blood, hair, skin cells, or other genetic evidences left at the scene of a crime can be compared through VNTR patterns, with the DNA of a criminal suspect to determine guilt or innocence. VNTR patterns are also useful in establishing the identity of a homicide victim, either from DNA found as evidence or from the body itself.

The Steps in DNA Fingerprinting technique (Fig. 5.16) is as follows:

1. **Extraction of DNA**
   The process of DNA fingerprinting starts with obtaining a sample of DNA from blood, semen, vaginal fluids, hair roots, teeth, bones, etc.

2. **Polymerase chain reaction (PCR)**
   In many situations, there is only a small amount of DNA available for DNA fingerprinting. If needed many copies of the DNA can be produced by PCR (DNA amplification).

3. **Fragmenting DNA**
   DNA is treated with restriction enzymes which cut the DNA into smaller fragments at specific sites.

4. **Separation of DNA by electrophoresis**
   During electrophoresis in an agarose gel, the DNA fragments are separated into bands of different sizes. The bands of separated DNA are sieved out of the gel using a nylon membrane (treated with chemicals that allow for it to break the hydrogen bonds of DNA so there are single strands).

5. **Denaturing DNA**
   The DNA on gels is denatured by using alkaline chemicals or by heating.

6. **Blotting**
   The DNA band pattern in the gel is transferred to a thin nylon membrane placed over the 'size fractionated DNA strand' by **Southern blotting**.

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